



Kingdom Of Bahrain
Ministry of Education
University of Bahrain
College Of Information And Technology



THAGHRAH:

Practical Cybersecurity Learning Platform Through Packet Analysis

Senior Project 2025-2026

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PROBLEM STATEMENT

1

Beginners understand networking theory but struggle with practical packet analysis

2

Real PCAP investigation can be overwhelming

3

Affects students' confidence when analyzing traffic investigations



OBJECTIVES

01

Improve
practical packet analysis skills
(eg.Wireshark)

02

Provide guided progressive
cybersecurity learning

03

Increase engagement
through gamification

04

Generate dynamic
AI-based investigation scenarios

05

Create a reusable
educational training environment

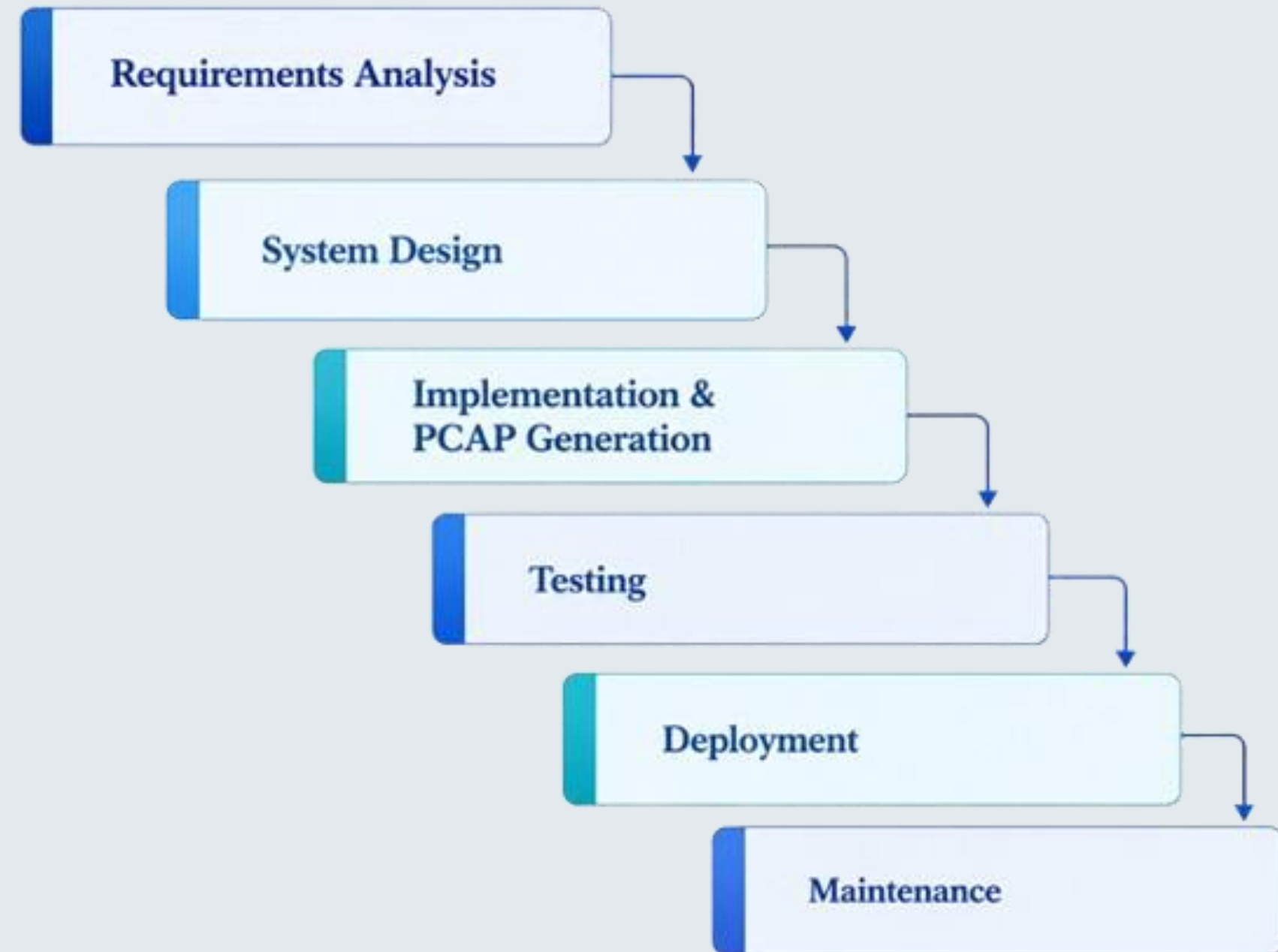
RELATED WORKS



PLATFORM	STRENGTH
OverTheWire	Uses SSH covering several topics. Progressive, guided learning including command-line.
TryHackMe	Offers training rooms, Guided learning using Wireshark.
Burp Academy	Specialized in HTTP/HTTPS, Monitor vulnerabilities and intercept requests via proxy monitoring.

Overall, these platforms offer great practical training, but they vary in their support for protocol-diverse traffic analysis and guided workflows.

SOFTWARE DEVELOPMENT LIFE CYCLE : WATERFALL MODEL



REQUIREMENT COLLECTION

SOURCE	KEY FINDINGS
Literature Review	Manual creation of realistic traffic is time consuming and difficult to maintain
Interviews: Rashid Alnoaimi (Cybersecurity Specialist) and Anonymous Participant	• Strong theoretical understanding but limited practicality • Limited hands-on reduces confidence in real scenarios • Guidance improves understanding of complex tasks • Gamification increases motivation and participation

FUNCTIONAL REQUIREMENTS

Dynamic PCAP Capture Environment

Sequential Challenge Unlocking

Flag Validation

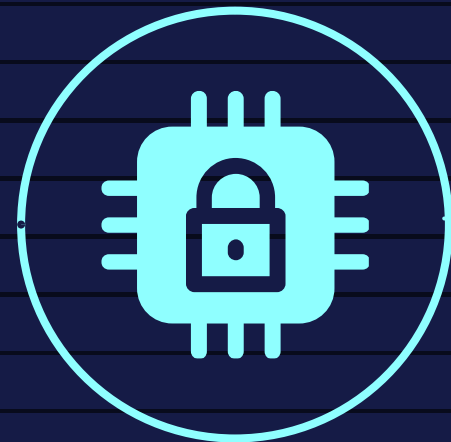
Protocol Guides & Cheat Sheets

Progress Tracking for Scoring & Badges

AI Challenge Generation

NON-FUNCTIONAL REQUIREMENTS

SECURITY



PORTABILITY

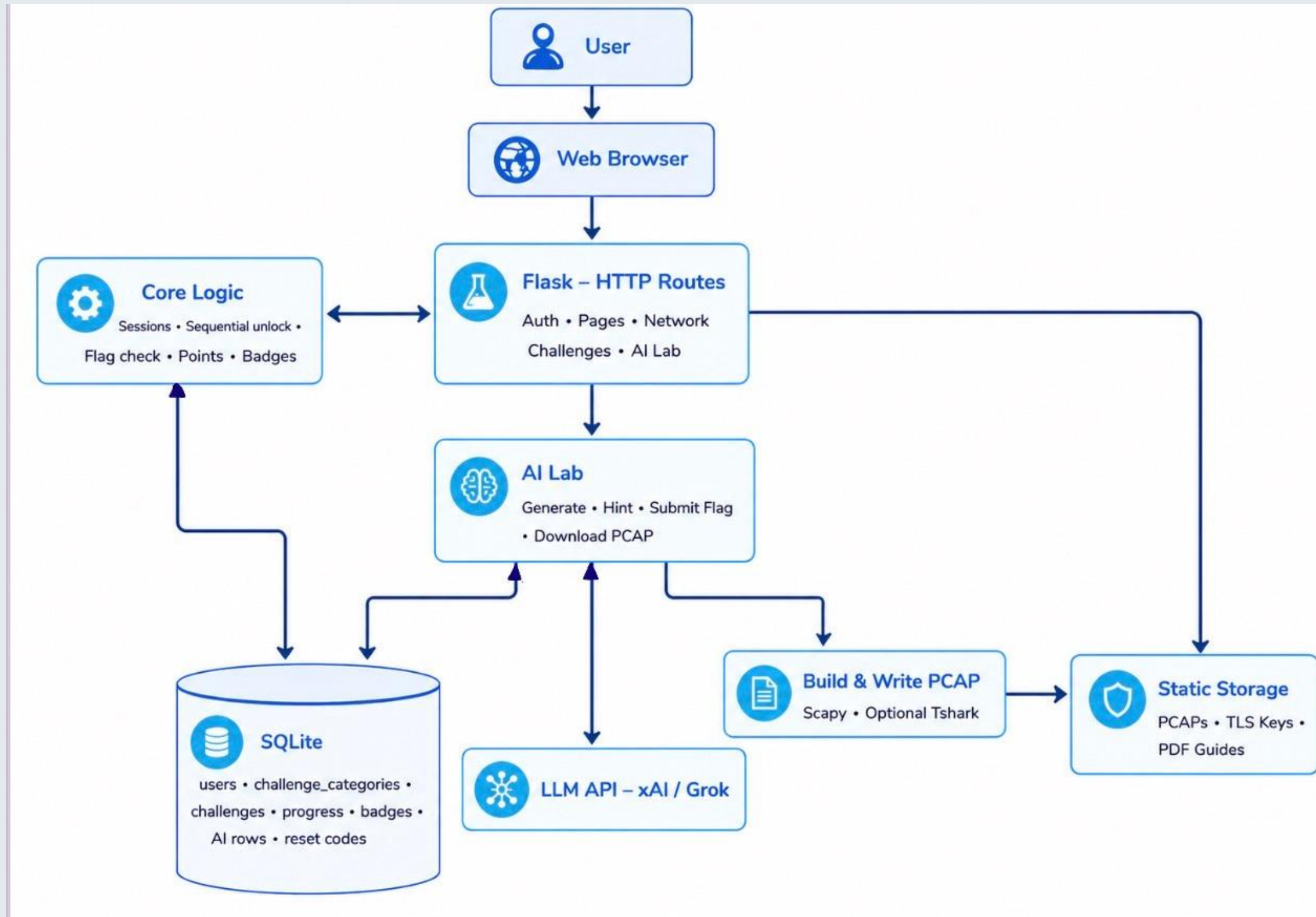
USABILITY



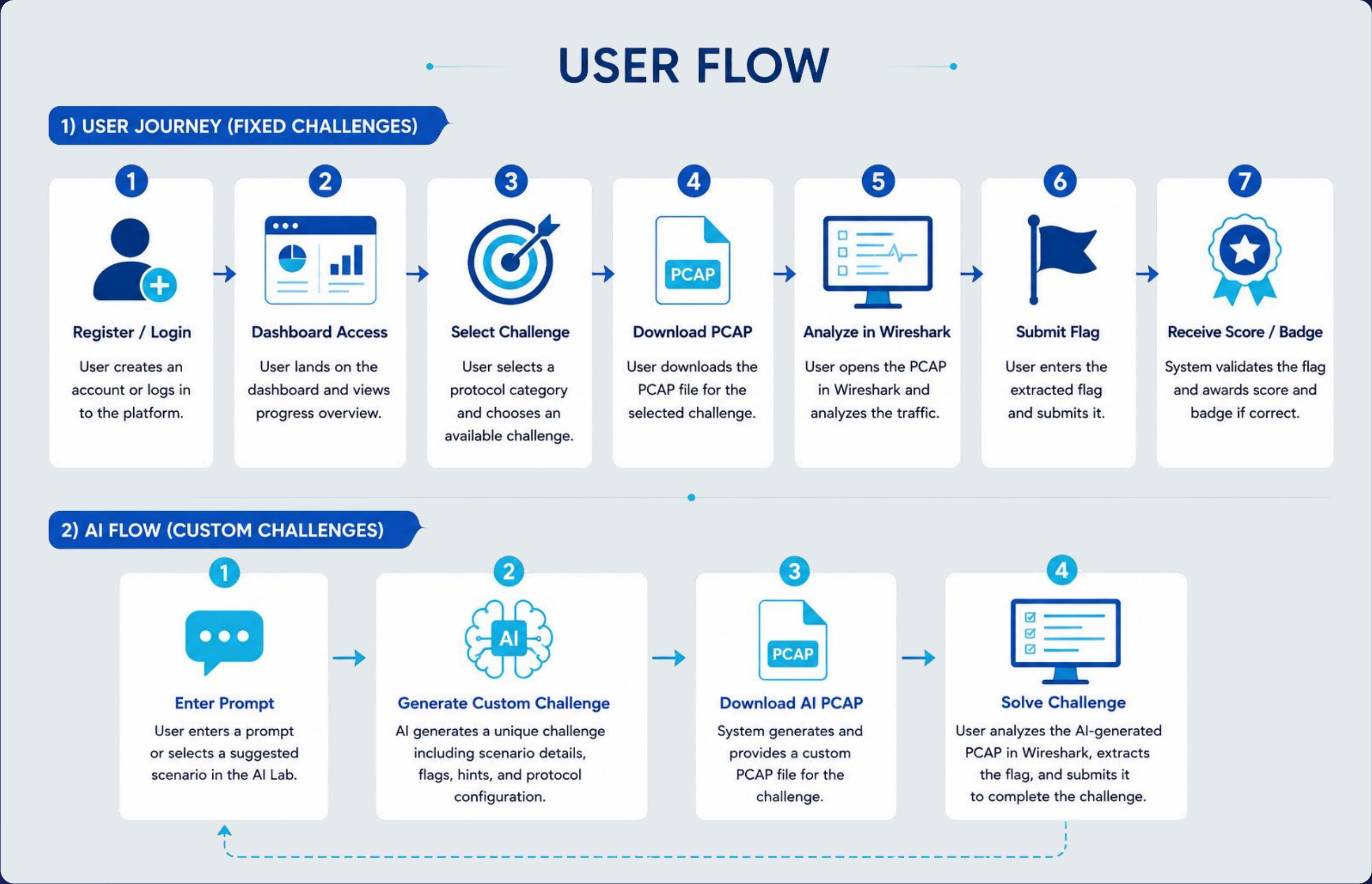
MAINTAINABILITY

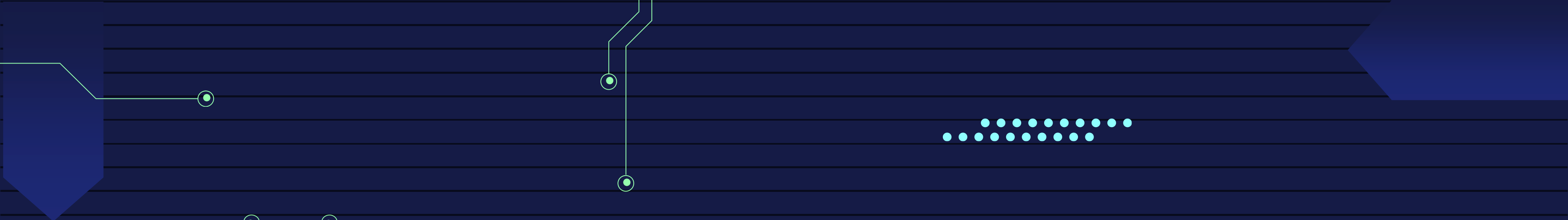


SYSTEM ARCHITECTURE



USERFLOW DIAGRAM





SYSTEM



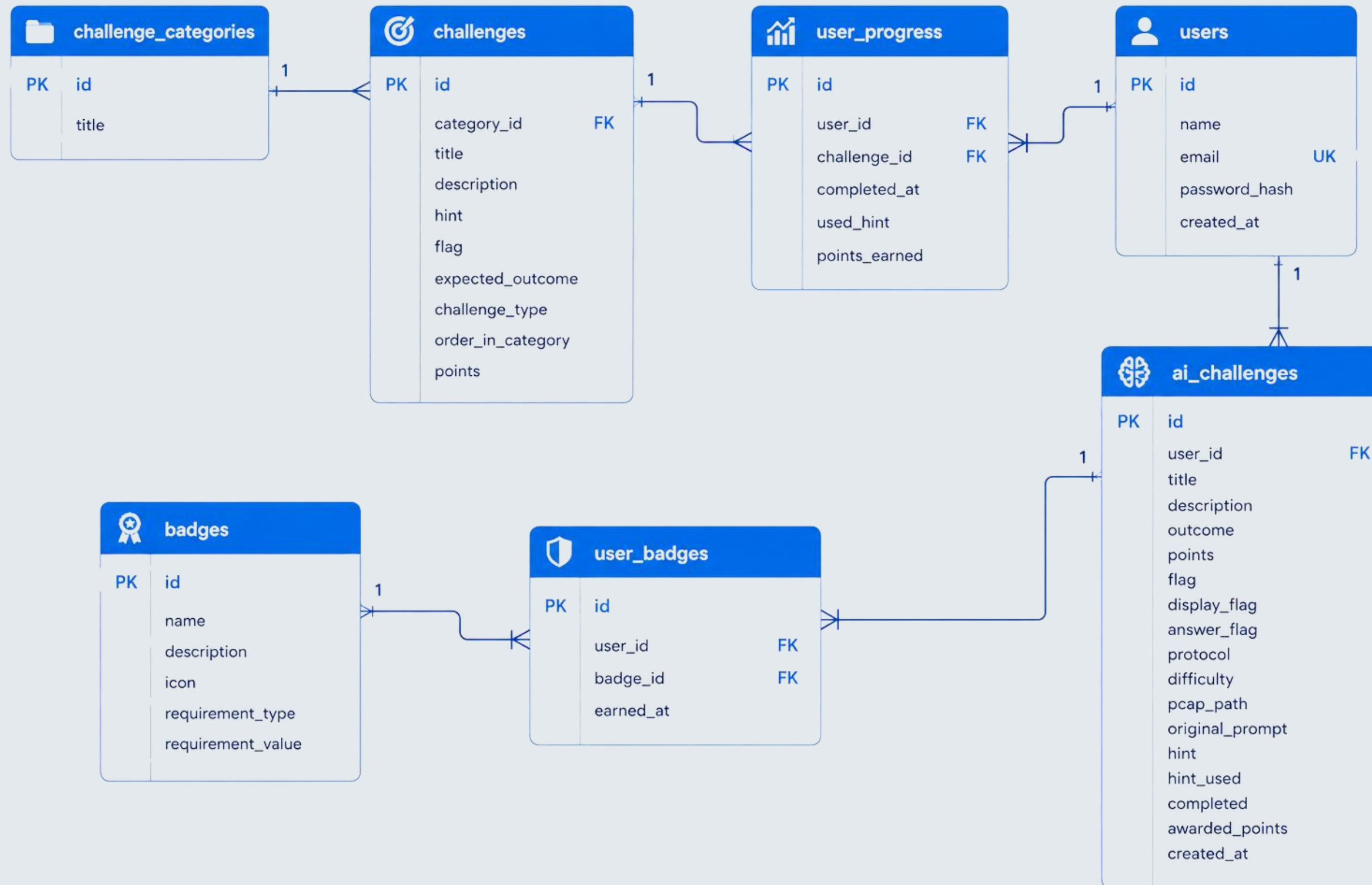
DEMONSTRATION



CHALLENGE 1:

TLS KEY

ENTITY RELATIONSHIP DIAGRAM



IMPLEMENTATION TECHNOLOGIES

BACKEND	1- Flask 2- Python
DATABASE	SQLite
PACKET GENERATION	1- Scapy 2- tshark(optional)
AI GENERATION	Grok API: xai Grok/Groq
SECURITY	Werkzeug password hashing
FRONTEND	1- HTML 2- CSS 3-JavaScript
ANALYSIS TOOL	Wireshark



IMPLEMENTATION TESTING

FUNCTIONAL TESTING

FUNCTIONAL AREA	DESCRIPTION	EXPECTED RESULT	STATUS
PCAP Download	Download the PCAP file associated with a challenge	Correct PCAP file downloads successfully	Pass
Flag Validation	Submit incorrect and correct flags for a challenge	Incorrect flags rejected; correct flags accepted with validation message	Pass
Hint Penalty	Open challenge hint and submit correct flag	Reduced score awarded after hint usage	Pass
Badge Awarding	Complete required challenge milestone	Badge notification displayed and reward assigned	Pass
AI Challenge Generation	Generate AI-based packet-analysis challenge from user prompt	Valid challenge, PCAP file, hint, and flag validation generated according to prompt	Pass

SECURITY TESTING

SECURITY AREA	DESCRIPTION	EXPECTED RESULT	STATUS
Password Hashing	Verify password storage in database and logs	Passwords stored and logged as hashes only	Pass
XSS Protection	Inject JavaScript payload in registration name field	Input treated as plain text without execution	Pass
Locked Challenge Access	Manually access locked challenge URL	User redirected to allowed challenge page	Pass
Authorization & Validation	Submit invalid or weak login credentials	Validation message displayed for login rejected and weak password	Pass

AI ACCURACY TESTING

Try No.	Prompt	Result	Description
1	DNS traffic containing Base64-encoded employee credentials hidden inside outbound queries	Pass	Accurate challenge generation with meaningful flag and valid PCAP workflow
2	ICMP traffic containing fragmented payloads that reconstruct hidden administrative credentials	Pass	Generated challenge and flag matched the supplied scenario correctly
3	SMTP mail server configured with vulnerable encryption settings allowing interception of confidential emails	Pass	Accurate scenario generation with coherent flag and downloadable PCAP
4	HTTP login portal targeted by repeated brute-force authentication attempts against administrator accounts	Pass	Challenge behavior and generated flag aligned with the prompt
5	FTP traffic with Base64-encoded authentication tokens transferred between client and server	Partial Pass	Challenge generated successfully, but flag value contained unrealistic random wording
6	Fragmented WhatsApp-style mobile traffic revealing hidden exfiltration behavior	Partial Pass	Generated challenge functioned correctly, but flag value lacked semantic meaning
7	FTP forensic challenge involving credential brute-force attacks against an exposed FTP server	Pass	Accurate challenge generation with valid flag and PCAP workflow
8	Medium-difficulty malware communication challenge using encoded HTTP traffic	Partial Pass	Scenario generated correctly, but generated flag contained a missing character
9	TLS traffic challenge involving fragmented Base64 credential transfer	Pass	Generated challenge and validation workflow behaved correctly
10	Brute-force login attempts against a target web application	Pass	Accurate challenge generation with coherent flag and downloadable PCAP

CONCLUSION & FUTURE WORK

To conclude, a practical, reusable cybersecurity learning environment for packet analysis training was created

Additional protocols
like (malware
analysis)

Increase the AI
reliability

Enhance gamification
features with
leaderboards

Live packets captures

Upgrade database
architecture



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THANK YOU

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ANY QUESTION?